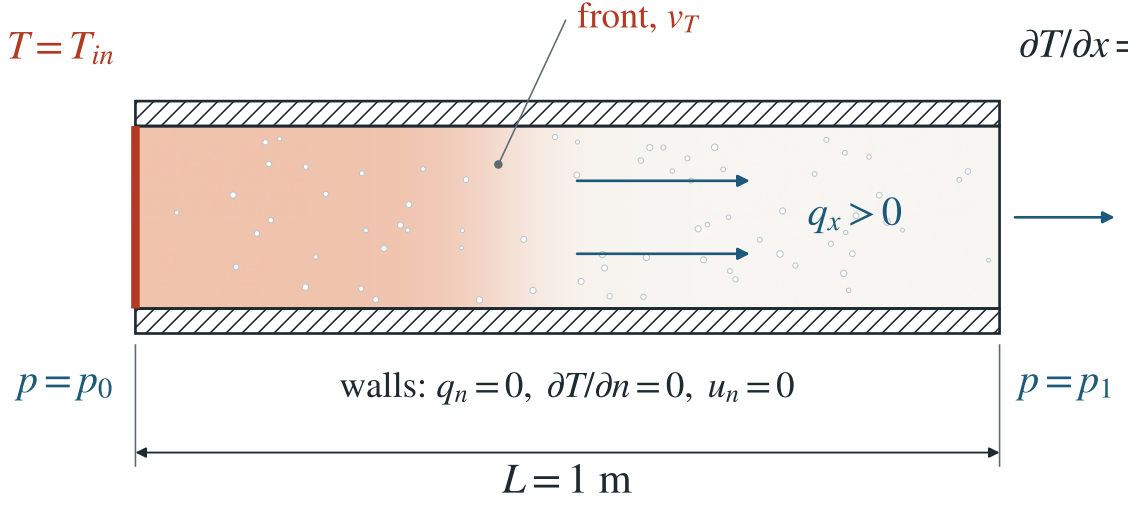
front, v_T
$T = T_{in}$
$\partial T/\partial x = 0$
$q_x > 0$
$p = p_0$
walls: $q_n = 0,\ \partial T/\partial n = 0,\ u_n = 0$
$p = p_1$
$L = 1$ m
initially $T = T_0;\ g = 0$